

CONFIDENTIAL
FOR OFFICIAL USE ONLY

NOT TO BE TAKEN INTO FRONT LINE TRENCHES

USE OF MINES IN TRENCH WARFARE

(From the French School of St. Cyr)

**TRANSLATED AND EDITED AT THE
ARMY WAR COLLEGE**

JULY, 1917



**WASHINGTON
GOVERNMENT PRINTING OFFICE
1917**

CONFIDENTIAL
FOR OFFICIAL USE ONLY

NOT TO BE TAKEN INTO FRONT LINE TRENCHES

USE OF MINES IN TRENCH WARFARE

(From the French School of St. Cyr)

TRANSLATED AND EDITED AT THE
ARMY WAR COLLEGE

JULY, 1917



WASHINGTON
GOVERNMENT PRINTING OFFICE
1917

The Project Gutenberg eBook of Use of Mines in Trench Warfare (From the French School of St. Cyr)

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Use of Mines in Trench Warfare (From the French School of St. Cyr)

Creator: Army War College

Release date: May 2, 2019 [eBook #59420]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59420

Credits: Produced by Brian Coe, John Campbell and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK USE OF MINES
IN TRENCH WARFARE (FROM THE FRENCH SCHOOL OF ST. CYR)

CONFIDENTIAL
FOR OFFICIAL USE ONLY

NOT TO BE TAKEN INTO FRONT LINE TRENCHES

USE OF MINES IN TRENCH WARFARE

(From the French School of St. Cyr)

TRANSLATED AND EDITED AT THE
ARMY WAR COLLEGE

JULY, 1917



WASHINGTON
GOVERNMENT PRINTING OFFICE
1917

WAR DEPARTMENT.

Document No. 635.

Office of The Adjutant General.

WAR DEPARTMENT,
WASHINGTON, *July 23, 1917.*

The following notes on Use of Mines in Trench Warfare are published for the Information of all concerned.

[062.1, A. G. O.]

BY ORDER OF THE SECRETARY OF WAR:

TASKER H. BLISS,
Major General, Acting Chief of Staff.

OFFICIAL:

H. P. McCAIN,
The Adjutant General.

WAR DEPARTMENT,
THE ADJUTANT GENERAL'S OFFICE ,
Washington, June 19, 1917.

To all officers of the Army:

You are advised that this and all subsequent documents of a similar character which may be furnished to you from this office are to be regarded as strictly confidential. They are to be kept at all times in your personal possession, and are not to be copied, nor are any parts of their contents to be communicated either directly or indirectly to the press, nor to any person not in the military or naval service of the United States. In Europe these documents are not to be carried into the front-line trenches, nor farther to the front than the usual post of the officers to whom issued.

Strict compliance with this injunction is enjoined upon every officer into whose hands any of these confidential documents may come.

BY ORDER OF THE SECRETARY OF WAR:

H. P. McCAIN,
The Adjutant General.

CONTENTS.

	Page.
Use of mines	5
Specially menaced points	5
Surface observations	5
Indications revealing enemy galleries	6
Listening for underground noise	6
Position of listening posts	6
Hours and precautions	7
What is heard	7
Distance at which noise can be heard	7
Direction from which noise comes	8
Useful information on mines	8
Defensive systems	8
Interval between galleries	8
Start, depth, and progress of work	9
Barrage trench for miners	9
Camouflet	9
Offensive galleries	10
Superimposed galleries	10
Mine chamber	11
Launching an attack	11
Craters	11
Craters to be occupied	11
Craters that the enemy occupies	12
Craters not occupied	12
Diagrams	9 , 10 , 12 , 13

USE OF MINES IN TRENCH WARFARE.

[From the French St. Cyr School.]

(Secret and Confidential. For Official Use Only. Not to be Taken into First-Line Trenches.)

USE OF MINES.

In sectors where the distance between the two lines is below 150 meters, mine warfare must be used. When the trenches are farther apart, underground warfare is seldom employed. In special cases, however, when there are strong ventilators and the line is stable enough to permit of it, advance may be made underground.

SPECIALLY MENACED POINTS.

The most vulnerable points evidently are the following: The outposts in advance of the line, machine-gun positions approximately located by the enemy, and the junction points of the communicating trenches with the first line.

SURFACE OBSERVATION.

Underground activity, either offensive or defensive, is first observed from those points in our lines nearest to it. All enemy trenches facing a salient of our lines will be the object of particular attention and closest daily observation. This observation of the first-line trenches should disclose the presence of enemy underground works and their approximate location.

One of the difficult questions in mining is the removal of the earth. Expert miners sometimes remove the earth as far as 100 to 200 meters from the entrance to the gallery. They throw it on old ruined shelters, in shell holes, on the reverse of the trenches. But these precautions are not always rigidly observed. When the noncommissioned officer is absent, or the enemy bombards a little strongly, some one in the working party not wanting to work overtime throws several clods of earth on the parapet.

INDICATIONS REVEALING THE ENEMY GALLERIES.

Freshly placed earth coming from underground is white and less dull in color than that of the parapets, which have been washed by the rain and blackened by explosives. The difference in color for the first day or so is striking. In chalk, large white spots are seen on the reverse of the German trench, daily growing larger. Without doubt they are working near by, and should be watched.

A communication trench comes out at a salient, and runs back from 150 to 200 meters. Patches of chalk, freshly moved and increasing daily, are observed. These are indications of underground work starting from the salient.

Four or five meters of enemy trench without loopholes, but with loopholes close together to the left and right, may mark a gallery entrance.

A miner's working relief reaches the gallery, each man carrying a piece of the frame or a lining plank over his shoulder, the ends of which can be seen over the parapet or through the loopholes. These are indications of a gallery position, especially if the men all move in the same direction and are lost to view at the same place.

From a raised point on the second or third line we see, with field glasses, an abnormal accumulation of sandbags in a well-known area. These sacks may indicate an underground gallery.

Patrols sent out in front of the enemy trenches sometimes bring back valuable information. They may hear the rumble of ventilators, the noise of a truck moving on the rails, men working near a gallery entrance, etc.

The enemy fire with heavy calibers on portions of the first line. Often the same corner is bombarded. This may indicate that the enemy artillery is seeking to facilitate the miner's work by overthrowing the entrances to troublesome galleries in the trench attacked.

Listening for underground noise.—The observation of hostile trenches may give indications which will limit the zone necessary to be watched; but for accurate results we must listen for underground noises.

Positions of listening posts.—The listening post is placed at the head of a gallery, in an angle of a gallery, in a deep dug-out, in a niche under the

parapet, or on the bottom of the trench. The points nearest to the enemy trench are selected for the listening posts. Below the outposts there is always a niche, allowing a man to place his ear to the ground (hence the name listening posts).

Hours and precautions.—The most favorable hours are morning about 4 or 5 o'clock, at 2 p. m., and at midnight. All the occupants of that part of the trench must keep still, and all work in the trench and the mine must cease for a given time.

What is heard.—The inexperienced ear hears too many things, and is easily mistaken in the noises heard. A relief passing in the enemy trench or in his own trench at 40 meters sounds strangely like the noise of a pick. A man hitting a ground sill or striking it with his heel gives the idea that work is being done. The impact of bullets on the parapet at night, when a fusillade is uninterrupted, also gives the idea of underground work. A man filing a fuse at the foot of his loophole suggests the presence of an enemy revolving borer. A man who snores beside the gallery entrance imitates the noise of a ventilator and may be mistaken for it.

However, to even a partially trained ear the noise of the pickax is characteristic. It is not a harsh sound, like that of a heel striking on chalk ground, nor is it like the shock of bullets piercing the parapet. It is a low, rhythmic sound, with regular cadence. In a gallery the miner works kneeling. When he has struck five or six blows with the pickax, he takes a breath. He repeats this process about 12 times. He stops two or three minutes, and the second miner clears away the earth and fills the truck with sacks. The first miner resumes his work. It is easy to distinguish this regular cadence peculiar to the miner.

In an infantry company there are always several miners by occupation, or several men familiar with engineering. These men are selected specially for the listening service. They can give accurate information to the officers and noncommissioned officers of a sapper company. It is also a good thing for a platoon commander to descend into a gallery and train his ear by exchanging pickax signals with the miner in the adjoining gallery.

Distance at which noise can be heard.—The following indications help in determining the distance of underground work:

Four men work in a gallery. They start the work, then stop. The ear is placed against the side of the wall, the other ear being covered by the hand. If the work is heard under these conditions, it is at a distance of 25 to 30 meters.

If all noise is avoided, and the work is heard without placing the ear against the wall, the distance is 12 to 15 meters.

If there is talking or working going on, and still the underground work is heard in spite of it, the distance is 8 to 10 meters.

At six meters a man can hear all the sounds of the pickax, the chalk crumbling, the pieces rolling down on one another, the sliding and stacking of cases. These noises sound as if they were immediately below.

At four meters a man can hear talking, the scraping of buttons against the wall, the miner turning around.

The humming of a ventilator can be heard at 40 meters without taking precautions to hear it.

An automatic borer can be heard all through the sector.

Directions from which noise comes.—It is easier to determine the direction of noise than the distance. There is always a chance of making observations in the galleries—one on the right, the other on the left of the noise. The exact location of the enemy underground work can be determined by intersections.

The engineer companies have an apparatus for intensifying the sound (strong microphones). They reinforce the sound when the apparatus is in the direction of the source of the sound. Their greatest defect is that they magnify sound too much and too many things are heard. Why hear for a distance of 100 meters when the enemy trench is only 40 meters away? Everything is heard in a mine gallery. It is difficult to distinguish among the many noises that of the enemy miner's pick. The ear is amply sufficient.

The beginner has a tendency to exaggerate the proximity of sounds. He thinks he is close to the enemy when he is still at a distance, and he takes steps to catch the enemy by exploding a camouflet, whose only effect is to retard his own work.

USEFUL INFORMATION ON MINES.

Defensive system.—Two arrangements can be adopted: Fan-shaped arrangement (Fig. A) or arrangement of independent parallel galleries (Fig. B). The second arrangement is preferable.

Interval between galleries.—Arrangement 2 being adopted, calculate the interval between two neighboring galleries in such manner as to prevent the enemy working underground. At the head of each gallery two elbows of 6 meters, with boring chambers, are made. A 6-meter boring is made from each chamber. At the bottom of the boring a camouflet is placed, effective for radius of 6 meters (not more heavily charged or the gallery will be demolished). The camouflet of the neighboring gallery forms a tangent to the first. The interval between the two galleries can not, therefore, be greater than 36 meters. In practice we would take 30 meters.

Start, depth, and progress of work.—Start from first line. Start at 4 meters, with a slope of 20 per cent to 30 per cent down to a depth of 10 meters. Then horizontal. Length to gallery and return without ventilation can cover 30 meters.

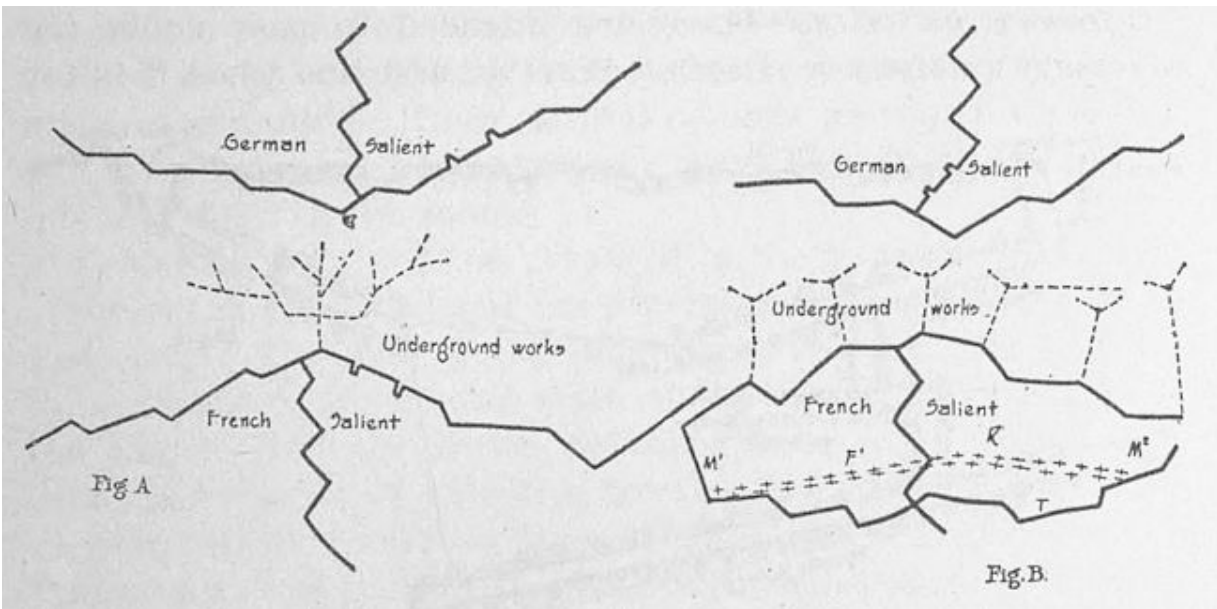


Fig. A.

Fig. B.

Barrage trench for miners.—If the enemy passes in spite of everything, the explosion should at least have been foreseen. The enemy's attack must

be limited or stopped, and this is always possible after the explosion of the charge, which may explode well in advance of our lines and act only as a strong means of launching his attack. The barrage trench is established at from 40 to 50 meters in rear of the salient T (Fig. B). In front of the parapet T, wire entanglement R, and two machine-gun positions M¹ and M² are placed. When the enemy's explosion is near, only a few men are left to occupy the salient. The German explosion does not bury anyone, and when the attack is launched it breaks down at the entanglement R.

Camouflet.—The sector commander is warned when the defense is about to explode a camouflet. About 10 meters on each side of the gallery are evacuated as a precaution. The only danger is in having several sandbags fall on the sides of the gallery entrance. Warn the working party charged with the relief, in order that they may not block the passage of men in the first line. Do not fire or make any changes in that part of the trench, in order that the enemy may not locate the position. Do not fire rockets before a given time. The camouflet sometimes shakes the ground and dust is visible. Nine times out of ten the camouflet is used at night.

When the enemy explodes a camouflet, fire a quantity of rockets to locate the positions. Fire grenades and throw bombs at the presumable position of the gallery. Send several men in front of the parapet to listen to what is going on in the enemy trench.

Offensive galleries.—These are intended to pass under the adversary's defenses. Depth.—Start at first line when it is far enough away from the enemy. Start at second line, or at special communication trench about 20 meters in the rear of the first line, when the enemy trench is too near. (See preceding.) A depth of 15 to 18 meters should be attained. The work is done as in ordinary galleries.

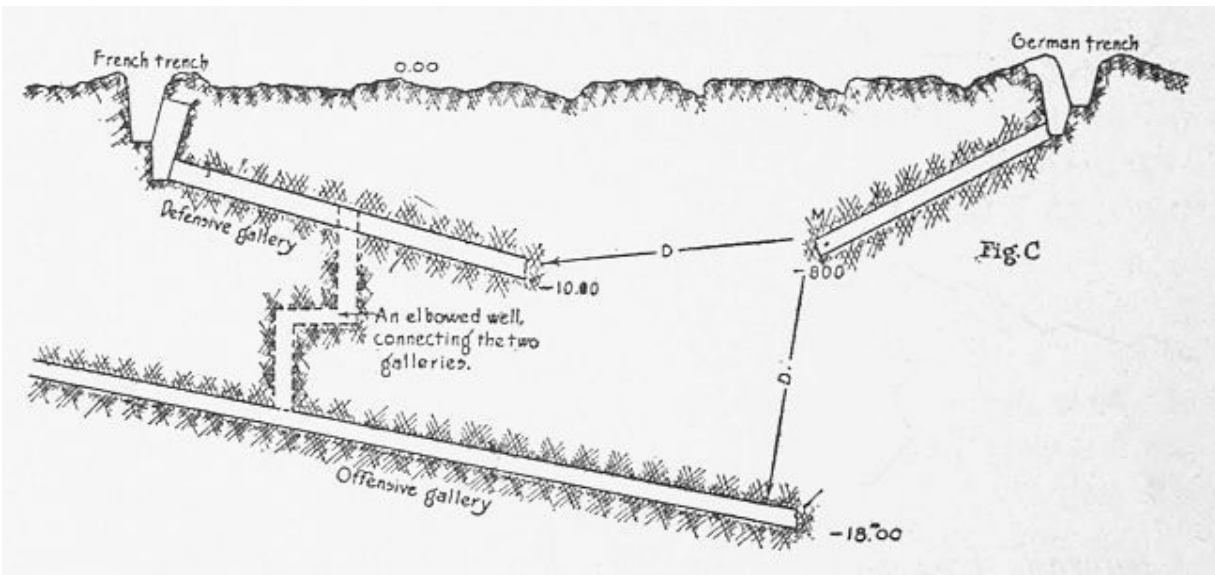


Fig. C.

Superimposed galleries.—This procedure gives splendid results in deceiving the enemy, who thinks he is protecting himself. The defensive gallery starts at the first line and the offensive gallery starts in rear of it. Both galleries are on the same vertical plane, the second being more advanced in the direction of the enemy than the first. An enemy listener easily confuses one with the other, and the offensive gallery passes under him. (See Fig. C.) Distances D and D' are the same. The miner, M , confuses the two sounds, and the offensive gallery passes under him.

Mine chambers.—They are of special interest to the Engineer Corps from a technical point of view. The best hours for exploding them are 4 a. m. and 7 p. m. When an attack is launched at a great distance, or when there is to be no attack, the explosion is preceded by several minutes of noise in the trenches. We commence firing and show several bayonets over the parapet. The enemy believes an attack is coming, comes out of his shelter, and mans his trenches. After the explosion we fire on the mine crater for four hours with artillery, grenades, and bombs. This fire should cover all the area exploded by the mine chamber and should prevent any help to the wounded or buried.

Launching an attack by means of mining.—The mine is an irresistible means of launching an attack. In a mined sector the best troops completely

lose their bearings for several seconds after an explosion. These several seconds prevent the machine gun from firing, and the assailant gains a foothold in the first line and often in the second.

A mine attack should be prepared in the following manner:

Several days beforehand the attacking troops are sent to the rest camp. The plan of the jumping-off trenches, the trenches to be attacked, the ground trace of the crater, and the zone of the searchlights are drawn on the ground with chalk. Each attacking fraction is placed in position, with the matériel to be carried. Each fraction's line of advance is marked out in chalk, as well as the section of the enemy's trench to be occupied and the position of the barrage. Every detail is studied thoroughly, and the exercise repeated a dozen times. The attack is then carried out as planned.

Craters.—The question of the occupation of craters must be decided by an authority higher than the platoon commander. In certain cases it is well to occupy them; in others, inadvisable. We will discuss only the practical work to be done in each case.

Craters to be occupied.—Crown the rim on the enemy side with a continuous trench, joined to the lines by (at least) two communicating trenches. Run out galleries in three directions, first for protection and later for use in the attack. Construct bombproofs on the half cone on the enemy side.

A crater is a position advantageous for the construction of auxiliary defenses, for the removal of earth from the galleries, for massing troops for an attack, and for flanking the lines.

Craters that the enemy occupies.—They should be hampered with bombs and grenades. We should head off the enemy by means of two well-placed mine chambers, which are always possible to fire rapidly when the scheme of defense is by independent galleries.

Craters not occupied.—It is well to see what goes on at the bottom. An outpost of several men or a sentinel may be posted for this purpose in a communicating trench on the friendly rim. Two communicating trenches may also be used to crown the friendly rim with a trench with slight counterslope. The outposts place a dozen loopholes permitting fire on the bottom. We may also fill up the bottom with chevaux de frise and other

auxiliary obstacles that can be thrown in. If the craters are in the way, we can always turn them by mining. (See figures for different examples:)

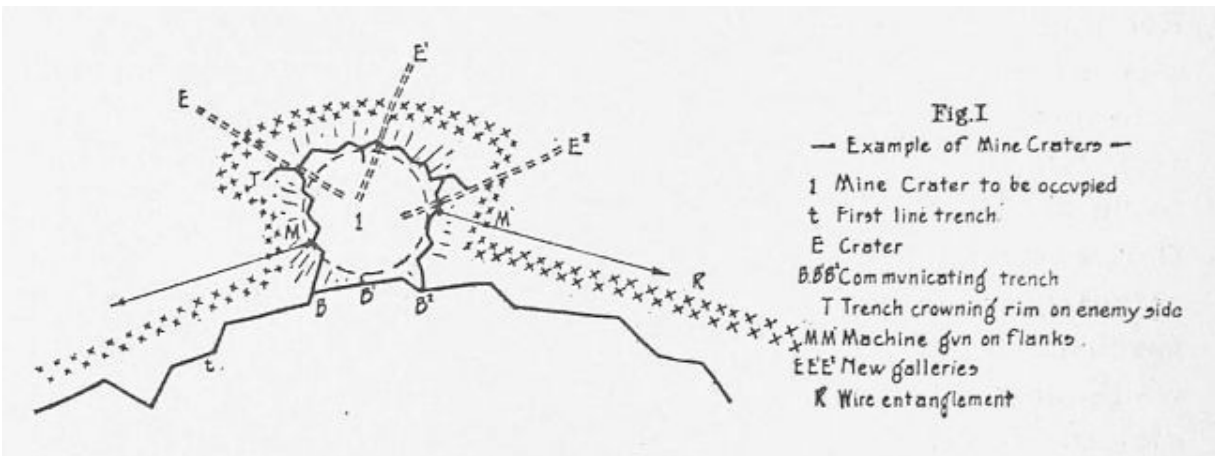


Fig. I.

—Example of Mine Craters—

- 1 Mine Crater to be occupied
- t First line trench
- E Crater
- B B' B'' Communicating trench
- T Trench crowning rim on enemy side
- M M' Machine gun on flanks
- E E' E'' New galleries
- R Wire entanglement

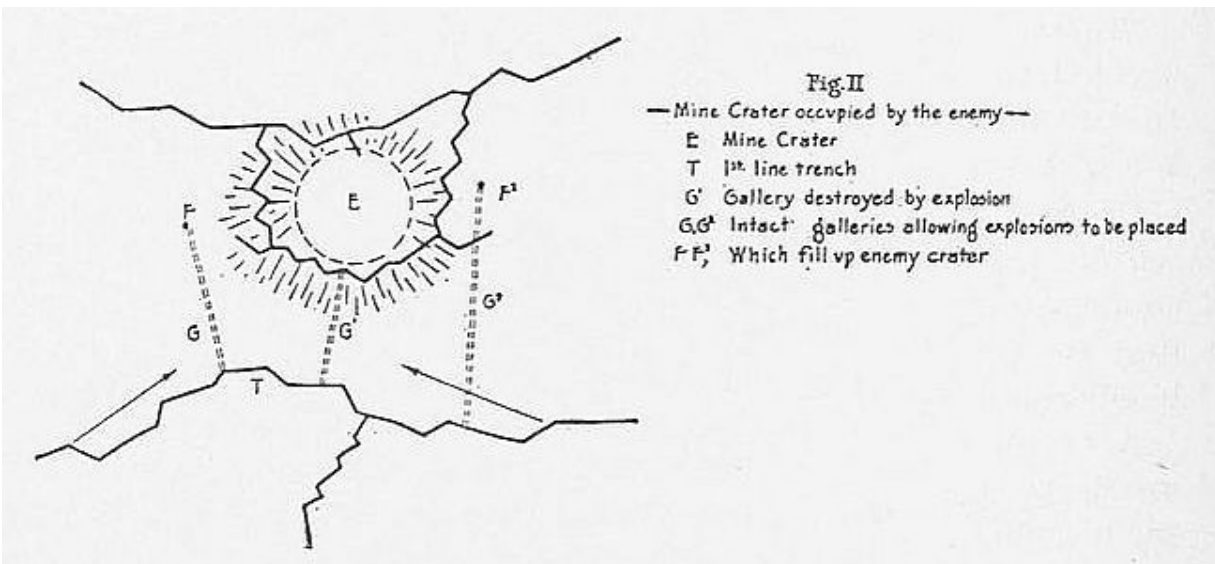


Fig. II.

—Mine Crater occupied by the enemy—

- E Mine Crater
- T 1st line trench
- G¹ Gallery destroyed by explosion
- G G² Intact galleries allowing explosions to be placed
- F F² Which fill up enemy crater

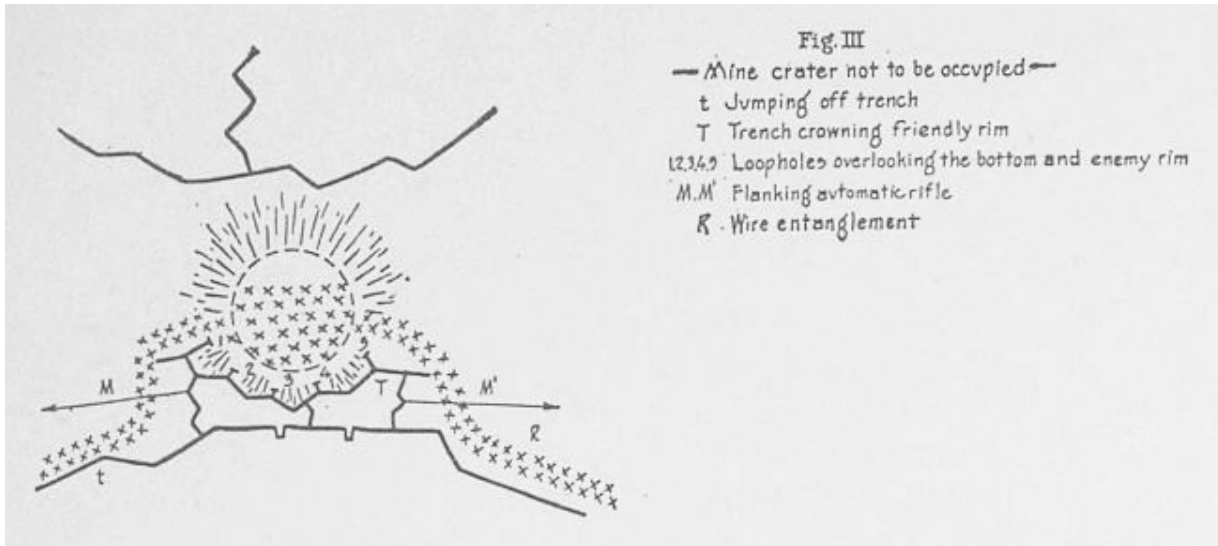


Fig. III.

—Mine crater not to be occupied—

- t Jumping off trench
- T Trench crowning friendly rim
- 1 2 3 4 5 Loopholes overlooking the bottom and enemy rim
- M M' Flanking automatic rifle
- R Wire entanglement

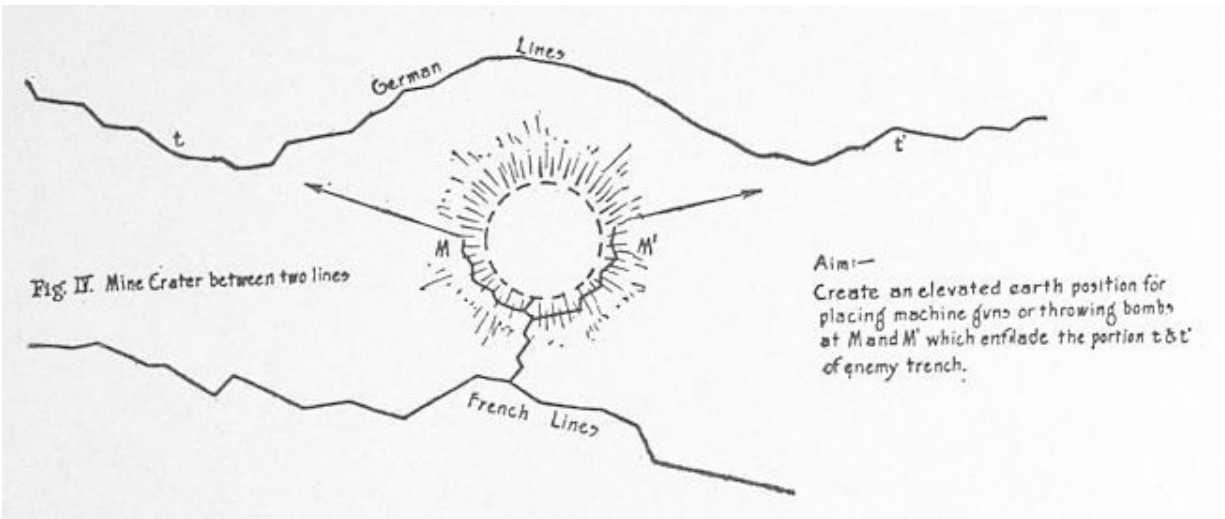


Fig. IV. Mine Crater between two lines

Aim:—
 Create an elevated earth position for
 placing machine guns or throwing bombs
 at M and M' which enfilade the portion t & t'
 of enemy trench.

Fig. V. Mine Crater for flanking purposes, see Fig. I & III.

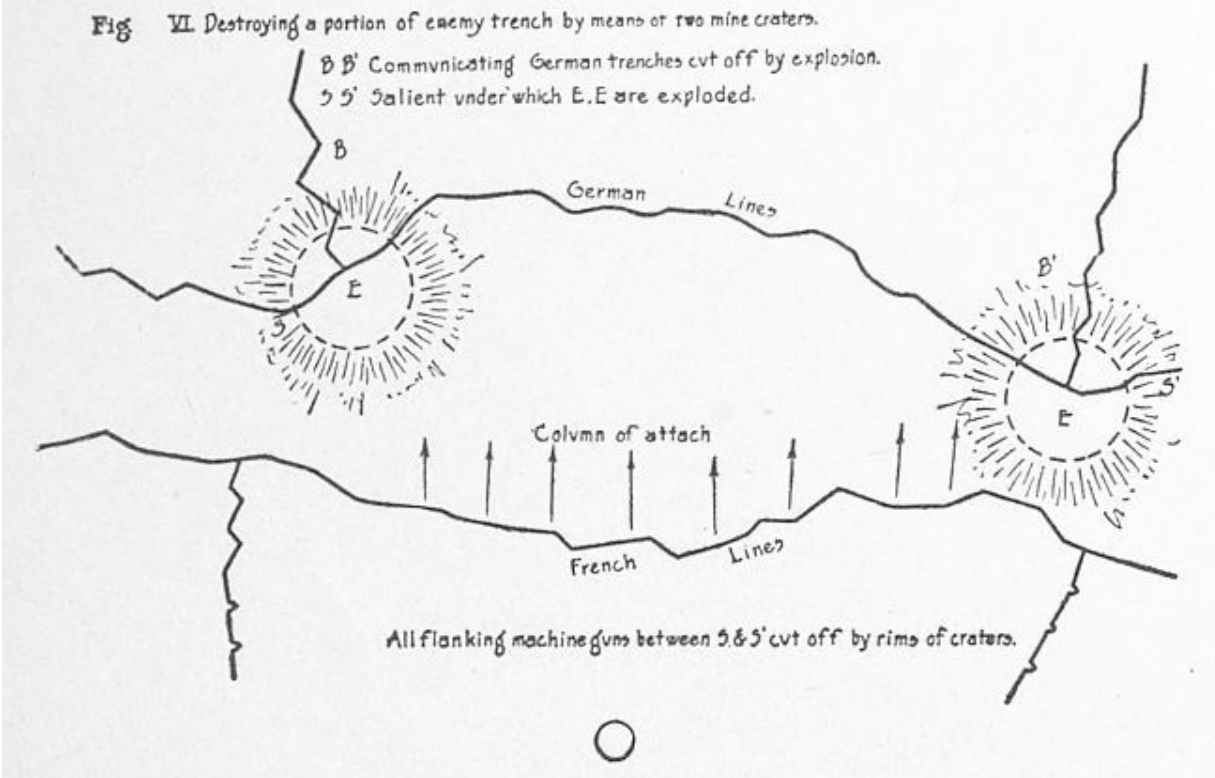


Fig. V. Mine Crater for flanking purposes, -1see Fig. I & III.

Fig. VI. Destroying a portion of enemy trench by means of two mine craters.

B B' Communicating German trenches cut off by explosion.

S S' Salient under which E.E are exploded.

All flanking machine guns between S & S' cut off by rims of craters.

TRANSCRIBER'S NOTE

Obvious typographical errors and punctuation errors have been corrected after careful comparison with other occurrences within the text and consultation of external sources.

Except for those changes noted below, all misspellings in the text, and inconsistent or archaic usage, have been retained.

[Pg 8](#): '(fig. a)' replaced by '(Fig. A)'.

[Pg 8](#): '(fig. b)' replaced by '(Fig. B)'.

[Pg 9](#): '(fig. 2)' replaced by '(Fig. B)'.

[Pg 9](#): 'positions M and' replaced by 'positions M¹ and'.

[Pg 10](#): '(See figure.)' replaced by '(See Fig. C.)'.

*** END OF THE PROJECT GUTENBERG EBOOK USE OF MINES IN
TRENCH WARFARE (FROM THE FRENCH SCHOOL OF ST. CYR) ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.